

Supplementary Information

August 19, 2025

The following derivations are presented as supplementary material to the paper **Active Learning Design: Modeling Force Output for Axisymmetric Soft Pneumatic Actuators**, authored by Gregory M. Campbell, Gentian Muhaxheri¹, Leonardo Ferreira Guilhoto, Christian D. Santangelo, Paris Perdikaris, James Pikul, and Mark Yim.

1 Theoretical actuator modeling

1.1 Elastomeric Thin Membrane

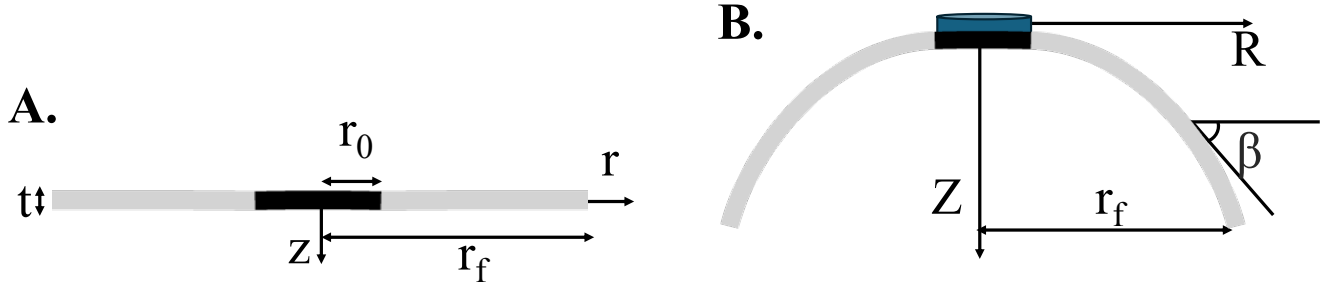


Figure 1: Schematic of the membrane (gray) with strain limiter (black) in the A. undeformed state, and B. deformed state while in contact with external force (blue).

We wish to determine the shape of a thin membrane upon inflation, with an external force applied at membrane radius, r , ranging from $0 \leq r \leq r_0$ as shown schematically in Fig. 1. Before deformation, the shape of the membrane that does not contact the force, using cylindrical coordinates, is given by $r_0 \leq r \leq r_f$, and membrane height $z = 0$. The region $0 \leq r \leq r_0$, which contacts the applied force, consists of low-strain material and we assume that there is no deformation in that region, only a translation of the region along the z direction. We assume the membrane stays axisymmetric after the deformation and its shape is given by,

$$R = R(r), \quad Z = Z(r). \quad (1)$$

From this, we can write the deformation gradient as,

$$\mathbf{F} = R'(r)\hat{\mathbf{r}} \otimes \hat{\mathbf{r}} + (R(r)/r)\hat{\theta} \otimes \hat{\theta} + Z'(r)\hat{\mathbf{z}} \otimes \hat{\mathbf{z}}, \quad (2)$$

where primed variables indicate a derivative with respect to r . The eigenvalues of the right Cauchy-Green tensor, namely of $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, are the principal stretches of the membrane,

$$\lambda_1(r) = \sqrt{R'(r)^2 + Z'(r)^2}, \quad \lambda_2(r) = \frac{R(r)}{r}, \quad (3)$$

where $\lambda_1(r)$ represents the amount of material stretch along the meridian direction, while $\lambda_2(r)$ represents the amount of material stretch along the latitude direction. The third principal stretch comes from our assumption of incompressibility, $\lambda_1(r)\lambda_2(r)\lambda_3(r) = 1 \rightarrow \lambda_3(r) = 1/(\lambda_1(r)\lambda_2(r))$, where $\lambda_3(r)$ represents the material stretch in the direction normal to the

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membrane (thickness direction). We write an energy for the system that assumes full contact between the contact plate and the strain limiting contact region as shown in Fig. 2B in the main text.

$$E = \int_{r_0}^{r_f} \left(\underbrace{2\pi r t W(\lambda_1, \lambda_2) dr}_{\text{elastic energy}} - \underbrace{\pi p R^2 Z' dr}_{\text{work done by pressure difference}} \right) + \int_0^{r_0} \left(\underbrace{-F Z' dr}_{\text{external force work}} + 2\pi r t \tilde{W}(\lambda_1, \lambda_2) dr - \pi p R^2 Z' dr \right), \quad (4)$$

where W and $\tilde{W}$ are the strain energy per unit undeformed volume for the elastic material and the low-strain material respectively, while p is the gauge pressure inflating the membrane. In this study, we use the Gent model for the strain energy density function [1], which we can write as $W = -\frac{\mu J_m}{2} \ln(1 - \frac{\lambda_1(r)^2 + \lambda_2(r)^2 + \lambda_3(r)^2 - 3}{J_m})$. This model contains two constants, the shear modulus μ and the extension limit constant J_m . We find the elastic material constants using uniaxial testing, similar to [2]: $\mu = 31.7 \text{ kPa}$, $J_m = 39.6$. We estimate sufficiently stiff values for the strain limiting material based on data in [3]: $\tilde{\mu} = 1 \text{ MPa}$, and $\tilde{J}_m = 25$.

We first set the first variation of the energy to zero,

$$\delta E = \int_{r_0}^{r_f} 2\pi r t \left(\frac{\partial W}{\partial \lambda_1} \frac{\partial \lambda_1}{\partial R'} \delta R' + \frac{\partial W}{\partial \lambda_2} \frac{\partial \lambda_2}{\partial Z'} \delta Z' \right) + \frac{\partial W}{\partial \lambda_2} \frac{\partial \lambda_2}{\partial R} \delta R - \int_{r_0}^{r_f} \pi p (2RZ' \delta R + R^2 \delta Z') dr + \int_0^{r_0} F \delta Z' = 0. \quad (5)$$

We then integrate by parts,

$$\delta E = \int_{r_0}^{r_f} dr \left(-\frac{d}{dr} \left(2\pi r t \frac{\partial W}{\partial \lambda_1} \frac{\partial \lambda_1}{\partial R'} \right) + 2\pi r t \frac{\partial W}{\partial \lambda_2} \frac{\partial \lambda_2}{\partial R} - 2\pi p R Z' \right) \delta R \quad (6)$$

$$+ \int_{r_0}^{r_f} dr \left(-\frac{d}{dr} \left(2\pi r t \frac{\partial W}{\partial \lambda_1} \frac{\partial \lambda_1}{\partial Z'} - \pi p R^2 \right) \right) \delta Z + \left(2\pi r t \frac{\partial W}{\partial \lambda_1} \frac{\partial \lambda_1}{\partial R'} \delta R \right) \Big|_{r_0}^{r_f} \quad (7)$$

$$+ \left(\left(2\pi r t \frac{\partial W}{\partial \lambda_1} \frac{\partial \lambda_1}{\partial Z'} - \pi p R^2 \right) \delta Z \right) \Big|_{r_0}^{r_f} + F \delta Z \Big|_0^{r_0} = 0. \quad (8)$$

Now, to obtain the equations of equilibrium, we set the 'Euler-Lagrange' equations obtained from above to zero,

$$-\frac{d}{dr} \left(2\pi r t \frac{\partial W}{\partial \lambda_1} \frac{\partial \lambda_1}{\partial R'} \right) + 2\pi r t \frac{\partial W}{\partial \lambda_2} \frac{\partial \lambda_2}{\partial R} - 2\pi p R Z' = 0 \quad (9)$$

$$-\frac{d}{dr} \left(2\pi r t \frac{\partial W}{\partial \lambda_1} \frac{\partial \lambda_1}{\partial Z'} - \pi p R^2 \right) = 0. \quad (10)$$

After taking the derivatives with respect to r and applying geometric relations based on $\beta(r)$, which we take to be the angle between the tangent to the membrane, $R'(r) = \lambda_1(r) \cos(\beta(r))$, $Z'(r) = \lambda_1(r) \sin(\beta(r))$, we write the equilibrium equations for the $r_0 \leq r \leq r_f$ region as:

$$\frac{d\lambda_1}{dr} = \frac{W_2 - \lambda_1 W_{12}}{r W_{11}} \cos \beta + \frac{\lambda_2 W_{12} - W_1}{r W_{11}}, \quad (11)$$

$$\frac{d\lambda_2}{dr} = \frac{\lambda_1 \cos \beta - \lambda_2}{r}, \quad (12)$$

$$\frac{d\beta}{dr} = \frac{\tilde{p} r \lambda_1 \lambda_2 - W_2 \sin \beta}{r W_1}, \quad (13)$$

where $\tilde{p} = p/t$, $W_1 = \partial W / \partial \lambda_1$, $W_2 = W / \partial \lambda_2$, and $W_{12} = \partial^2 W / \partial \lambda_1 \partial \lambda_2$.

The second term in Eq. 4 only contributes to the boundary conditions at r_0 since there is no deformation in the $0 \leq r < r_0$ region. To obtain the boundary condition for β at $r = r_0$, we use the boundary force terms in the z direction, at $r = r_0$, obtained from the integration by parts above,

$$-2\pi r t \frac{\partial W}{\partial \lambda_1} \frac{\partial \lambda_1}{\partial Z'} + \pi p R^2 - F = 0. \quad (14)$$

Solving this equation for $\beta(r_0)$ provides us with the necessary boundary condition. We can now write all three conditions needed to solve for the shape of the membrane,

$$\lambda_1(r_0) = x; \lambda_2(r_0) = 1; \beta(r_0) = \text{ArcSin} \left(\frac{\pi p r_0^2 \lambda_2^2 - F}{2\pi t r_0 W_1} \right) \Big|_{r=r_0} \quad (15)$$

The condition on λ_2 comes from the fact that the contact area and the plate are in full contact and there is no extension in that area. The condition on $\beta(r_0)$ comes from the boundary conditions obtained during energy minimization, namely $\left(-(2\pi r t \frac{\partial W}{\partial \lambda_1} \frac{\partial \lambda_1}{\partial Z'} - \pi p R^2) \delta Z \right) \Big|_{r=r_0} + F \delta Z \Big|_{r=r_0} = 0$. We provide values for p and F , and solve the condition on λ_1 by using the shooting method to find the value of x that satisfies the condition $\lambda_2(r_f) = 1$. This final condition states that the membrane is fixed at the ends. Integrating the equilibrium equations from r_0 to r_f obtains the shape of the deformed membrane, and thus we find the height of the membrane for a given pressure and force value. Solving this relationship across many pressure-force pairings creates a theoretical model that can estimate forces for pressure-height pairings within, or nearby, the set.

1.2 Concentrically Strain Limited Thin Membrane

To span our actuator design space, we want to find the deformed shape of a membrane made up of elastic material that also includes strain limiting rings (see Fig. 2 in the main paper). To do this, the energy in Eq. (4) will involve an extra integral for each additional piece we add to the system, with the strain energy density function W being used for the elastic material and $\tilde{W}$ being used for the strain limiting rings.

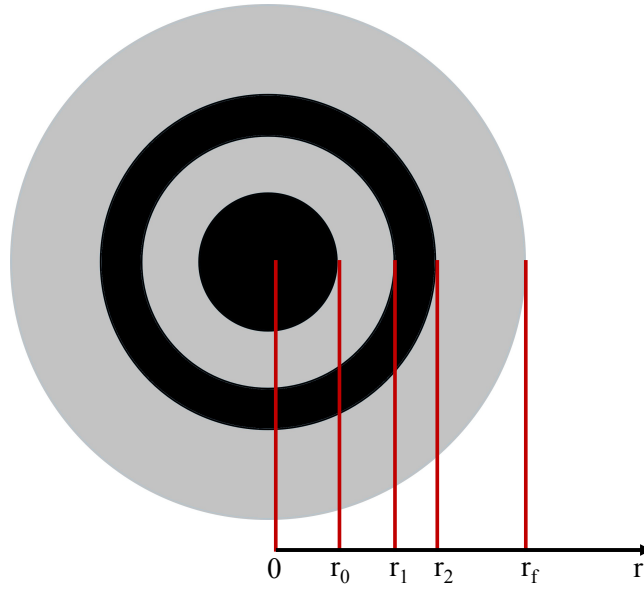


Figure 2: Schematic of the concentrically strain limited membrane where the strain limiters are depicted in black.

1.2.1 Example: Thin Membrane with Two Strain Limiters

To illustrate the procedure, we analyze a thin membrane that has one strain limiting region in addition to the contact area, which is also made up of low-strain material. For this system we can write the energy as,

$$\begin{aligned} E = & \int_0^{r_0} (-F Z' dr + 2\pi r t \tilde{W}(\lambda_1, \lambda_2) dr - \pi p R^2 Z' dr) \\ & + \int_{r_0}^{r_1} (2\pi r t W(\lambda_1^{(1)}, \lambda_2^{(1)}) dr - \pi p R^{2(1)} Z'^{(1)} dr) + \\ & + \int_{r_1}^{r_2} (2\pi r t \tilde{W}(\lambda_1^{(2)}, \lambda_2^{(2)}) dr - \pi p R^{2(2)} Z'^{(2)} dr) + \\ & + \int_{r_2}^{r_f} (2\pi r t W(\lambda_1^{(3)}, \lambda_2^{(3)}) dr - \pi p R^{2(3)} Z'^{(3)} dr), \end{aligned} \quad (16)$$

where $\tilde{W}$ is the strain energy density function for the low strain material, and the superscripts in parentheses differentiate the functions by region. The region involving $r = r_0$ to $r = r_1$ is region 1 and the rest follows. We proceed as in the section above, by taking the first variation of the energy functional and setting it to zero. After integrating by parts, we obtain the same equations of equilibrium as Eqs. [11-13], but we now have one set of those equations for each region of the membrane. That is, for this example, we have three sets of those equations. The boundary conditions at $r = r_0$ are once again identical to Eq. 17,

$$\lambda_1^{(1)}(r_0) = x; \lambda_2^{(1)}(r_0) = 1; \beta^{(1)}(r_0) = \text{ArcSin} \left(\frac{\pi p r_0^2 \lambda_2^{2(1)} - F}{2\pi t r_0 W_1^{(1)}} \right) \Big|_{r=r_0}. \quad (17)$$

The conditions for the other regions are found from the force boundary terms that are obtained through integration by parts as in the previous section, and they are

$$\frac{\partial W^{(2)}}{\partial \lambda_1^{(2)}} \Big|_{r=r_1} = \frac{\partial W^{(1)}}{\partial \lambda_1^{(1)}} \Big|_{r=r_1}; \lambda_2^{(2)}(r_1) = \lambda_2^{(1)}(r_1); \beta^{(2)}(r_1) = \beta^{(1)}(r_1), \quad (18)$$

where the first equation in Eq. 18 is used to solve for $\lambda_1^{(2)}(r_1)$. Similarly, the rest of the conditions needed are given by,

$$\frac{\partial W^{(3)}}{\partial \lambda_1^{(3)}} \Big|_{r=r_2} = \frac{\partial W^{(2)}}{\partial \lambda_1^{(2)}} \Big|_{r=r_2}; \lambda_2^{(3)}(r_2) = \lambda_2^{(2)}(r_2); \beta^{(3)}(r_2) = \beta^{(2)}(r_2), \quad (19)$$

and

$$\lambda_2^{(3)}(r_f) = 1, \quad (20)$$

where this last condition is used to find x . As can be seen, each consecutive boundary condition is dependent on the previous one, so as a result, all of them are conditioned on what $\lambda_1^{(1)}(r_0)$ is, which is conditioned on $\lambda_2^{(3)}(r_f)$ being equal to 1. As a result, all of the equations of equilibrium combined with the boundary conditions given by Eqs. [17-20] are solved simultaneously to obtain the full membrane shape.

References

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